

Zeyu Lai

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Education

University of California, San Diego	2025 – 2029
Undergraduate in Data Science & Electrical Engineering; Minor in Mathematics	
BASIS International School Park Lane Harbour	2022 – 2025
High School Diploma GPA: 3.79/4.0	

Experiences

Su Hao's AI Lab	Mar 2026 – Present
<i>Research Assistant (Embodied AI research group) UC San Diego</i>	
Developing scalable RL pipelines for robotic manipulation while leading hardware-in-the-loop integration and sim-to-real verification to facilitate high-fidelity data collection for complex dexterous tasks.	
MURO Lab	Oct 2025 – May 2026
<i>Research Assistant (Distributed Robotics and Multi-Agent Coordination) UC San Diego</i>	
Researched distributed robotics and multi-agent coordination; Developing multi-robot map-merging infrastructure to enable autonomous navigation for robot fleet (TurtleBot4 platforms) and unified it into a global representation.	
TRITON-AI	Oct 2025 – May 2026
<i>Path Planning Team Member (Autonomous Vehicles) UC San Diego</i>	
Worked on LiDAR-camera sensor fusion for F1TENTH and contributed to the path planning team for Triton-AI's autonomous go-kart.	

Projects

DPPO for Dexterous Manipulation	2026
Implemented Diffusion Policy Policy Optimization (DPPO) for high-dimensional, state-based dexterous manipulation tasks (Shadow Hand). Developed a modular pipeline featuring SAC-based expert data generation, action-chunking, state normalization, and bridged generative diffusion models with RL by fine-tuning a pre-trained diffusion policy to maximize task rewards using PPO.	
RL Gymnasium	2026
Developed a clean, extensible reinforcement learning repository for training and evaluating classic Gymnasium control tasks, utilizing Python, PyTorch, and reinforcement learning principles for algorithmic implementation.	
Uniq: Intelligent Browser Automation System	2026
Engineered an autonomous browser agent using Node.js, Playwright, and Gemini 3.0 Flash for task planning and decision making. Developed a resilient Agent Loop featuring semantic auto-scroll, graduated failure recovery, and DOM-based interactive region detection using TypeScript and WebSockets.	
Hybrid Crawler-Drone Robot System	2025
Designed and built a multi-modal vehicle with a detachable drone basket and retractable arms. Integrated ESP32-S3 with I2S microphone, SPIFFS file system, and motor drivers for ground motion, offering unique flexibility for real-world navigation and reducing costs compared to separate robots.	

English–Chinese NMT with Transformer

2024

Built a Transformer-based neural machine translation (NMT) model from scratch in PyTorch. Used Hugging Face’s opus-100 dataset for English–Chinese parallel sentence training, gaining hands-on experience with tokenizers, transformer architecture, model training, and debugging.

Gesture-Based Brightness Adjuster

2024

Created a webcam-controlled brightness tool using OpenCV and MediaPipe. Mapped thumb–index finger distance to screen brightness and used a pinky curl as a trigger for fixed levels. Won 1st out of 36 participants at the BASIS national hackathon. Served as a low-cost alternative to hardware controls, useful in environments where hygiene and contactless interaction are key.

Monte Carlo Ising Simulation

2024

Simulated 2D Ising spin-lattice models using Monte Carlo methods to study phase transitions and equilibrium states under Prof. Erik Luijten at Northwestern University (Cetus Research), utilizing Python, NumPy, and Matplotlib.

Smart Glasses for Visually Impaired (NOVA MAKER)

2023

Invented and developed AR smart glasses achieving 90% route detection accuracy with embedded text-to-audio translation. Field-tested with over 20 users and recognized as a National Finalist at the Invention Convention China 2024.

Publications

Diving into the virtual realm: Exploring the mechanics of virtual reality

[ResearchGate]

Zeyu Lai

Proc of the 2023 International Conference on Machine Learning and Automation

Technical Skills

Languages: Python, C, Java, JavaScript, Typescript

Libraries & Tools: NumPy, OpenCV, MediaPipe, ROS2, micro-ROS, SLAM Toolbox, Pytorch, Scikit-Learn, Isaac Gym, Isaac Lab, Isaac Sim, Gymnasium

Hardware: X-ARM7, LEAP HAND, ESP32, Arduino, Turtlebot4, Jetson Nano, Sensor integration (LiDAR, IMU, etc.)

Machine Learning & AI: Deep learning architectures, Reinforcement learning, Imitation Learning

Design & Dev Tools: PCB Design, Fusion 360, Jupyter notebook, Google Colab, Conda, GitHub Actions, Linux

Honors & Certifications

Honors:

- 2024 USACO Gold Division
- 2024 Invention Convention China National Finalist
- BASIS China Hackathon 1st Place (1/36)
- Euclid Math Contest Distinction (top 20%)
- IMMC 2024 Greater China Region Finalist

Certifications (Stanford University):

- Supervised ML: Regression & Classification
- Advanced Learning Algorithms
- Unsupervised Learning, Recommenders, RL